BANKNOTE AUTHENTICATION USING LOGISTIC REGRESSION

Machine Learning Mini-Project

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Course: Machine Learning

1 Problem Statement

The objective of this project is to develop a Logistic Regression classifier that predicts a banknote dataset's binary class label using four numerical measurements extracted from scanned banknote images.

The model is evaluated on unseen test data using suitable classification metrics and a confusion matrix.

This is an educational project and is not intended for real banking, legal, or security decisions.

2 Dataset Description

The Banknote Authentication Dataset from the UCI Machine Learning Repository contains 1,372 records, four numerical input features, and a binary target variable named `Class`.

The input features are computed from banknote images using a wavelet transform.

Table 1: Dataset Features

Feature	Description
Variance	Variance of the wavelet-transformed image
Skewness	Skewness of the transformed image
Curtosis	Kurtosis-related measurement
Entropy	Entropy of the image
Class	Binary target label: 0 or 1

The dataset contains 762 records in Class 0 and 610 records in Class 1. No missing values were found.

The meaning of each class label should be stated according to the official UCI dataset documentation.

3 Methodology

The dataset was loaded using Pandas and inspected for its dimensions, data types, missing values, summary statistics, and class distribution.

The four feature columns were assigned to the input matrix X, while the target column was assigned to y.

The dataset was divided into 80% training data and 20% testing data using a random state of 42 and stratified sampling.

A Scikit-learn pipeline was created using StandardScaler followed by Logistic Regression.

- StandardScaler standardizes the input features.
- Logistic Regression learns the relationship between the features and the target class.
- The scaler is fitted only on training data to prevent data leakage.

The trained model generated class predictions and probabilities for the test set.

4 Data Visualization

4.1 Class Distribution

Figure 1 shows the distribution of the two classes in the dataset.

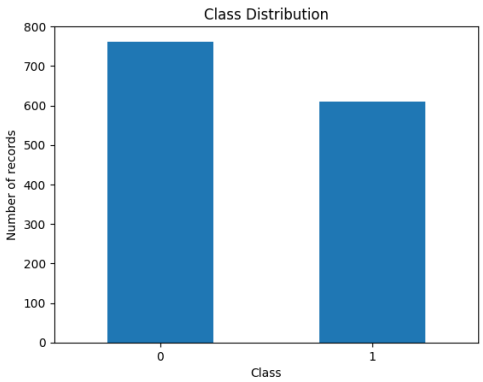


Figure 1: Class distribution of the Banknote Authentication Dataset

Observation: The dataset contains more Class 0 records than Class 1 records, although the difference is not large.

4.2 Feature Distributions

Figure 2 presents histograms of the four input features.

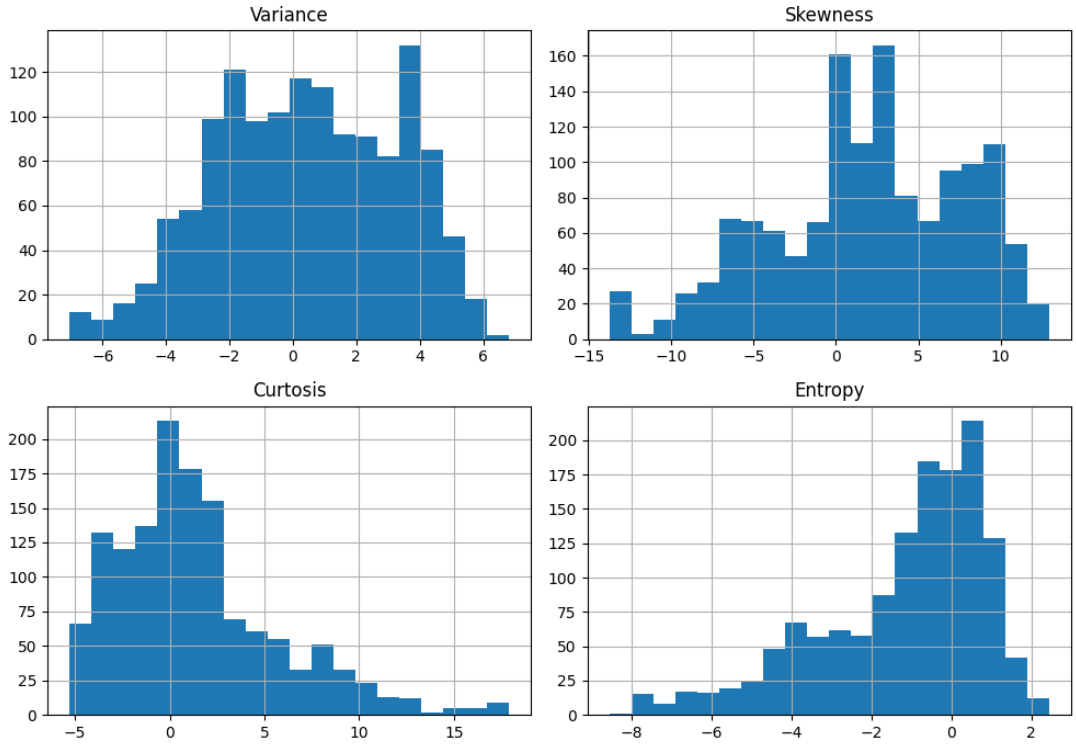


Figure 2: Distribution of the four numerical input features

Observations: The features have different numerical ranges and distributions. This supports the use of feature scaling before training Logistic Regression.

5 Model Evaluation

The trained model was evaluated using accuracy, precision, recall, and F1-score.

Table 2: Model Performance on Test Data

Metric	**Result**
Accuracy	97.09%
Precision (Class 1)	93.85%
Recall (Class 1)	100.00%
F1-score (Class 1)	96.83%

The model correctly classified 267 out of 275 test records.

5.1 Confusion Matrix

The confusion matrix is shown in Figure 3.

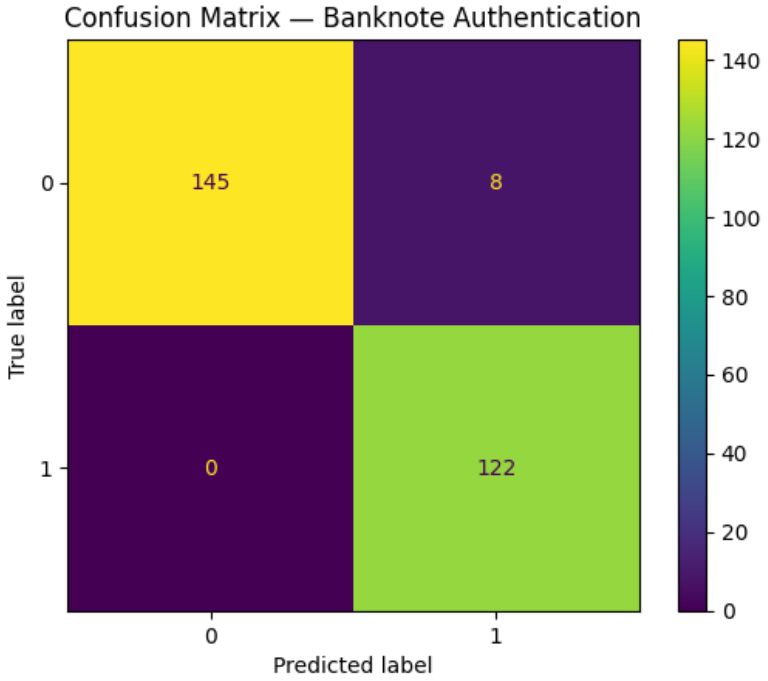


Figure 3: Confusion matrix for Logistic Regression

The confusion matrix is:

$$\begin{bmatrix} 145 & 8 \\ 0 & 122 \end{bmatrix}$$

The model produced 145 true negatives, 8 false positives, 0 false negatives, and 122 true positives.

6 Threshold Comparison

The classification threshold was compared at 0.30, 0.50, and 0.70 for predicting Class 1.

At thresholds 0.30 and 0.50, the model achieved an accuracy of 97.09%, precision of 93.85%, recall of 100%, and F1-score of 96.83%.

At threshold 0.70, accuracy increased to 99.64%, precision increased to 99.19%, recall remained at 100%, and F1-score increased to 99.59%.

The higher threshold reduced false-positive predictions without losing any Class 1 cases in this test set.

7 Interpretation

Regression predicts a numerical value, whereas classification predicts a category.

Logistic Regression is used for classification because it applies a sigmoid function to a linear score to estimate a probability between 0 and 1.

Accuracy measures the proportion of correct predictions. Precision measures how many predicted Class 1 cases are actually Class 1. Recall measures how many actual Class 1 cases are correctly identified. F1-score is the harmonic mean of precision and recall.

The importance of each metric depends on the cost of different classification errors and the verified meaning of the class labels.

8 Limitations

The project uses engineered numerical features rather than raw banknote images.

The dataset comes from a specific educational collection, so the results may not generalize to other data sources or real-world banknote screening conditions.

The model has not been validated for real banking or security use.

9 Conclusion

The Logistic Regression pipeline achieved 97.09% accuracy on the held-out test set, with 93.85% precision, 100% recall, and a 96.83% F1-score for Class 1.

The confusion matrix contained 145 true negatives, 8 false positives, 0 false negatives, and 122 true positives.

The threshold comparison showed that a threshold of 0.70 improved precision and accuracy on this test split while retaining 100% recall. However, threshold selection should depend on the relative costs of errors and should be validated rather than selected only from this test result.

A major limitation is that the model uses engineered features from one educational dataset and is not validated for real-world banking or security applications.

10 References

1. UCI Machine Learning Repository, *Banknote Authentication Dataset*.
   `https://archive.ics.uci.edu/dataset/267/banknote+authentication`

2. Scikit-learn Documentation, *Logistic Regression*.
   `https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression`